

Ball Machine

Madina yavumbuye ball machine kugira ngo ishimishe abitabiriye IOI. Internal structure ya machine imeze kuno:

- Iyi machine igizwe na N **nodes**, iri indexed kuva kuri 0 kugeza kuri $N - 1$. Node $N - 1$ yitwa **root**.
- Kuri buri node u ifite $0 \leq u < N - 1$, **parent** yiyo node u ni node $P[u]$ ifite $P[u] > u$, naho node u ni **child** ya $P[u]$. Root nta mubyeyi ifite. Node idafite children yitwa **leaf**.

Ntabwo uzi N cyangwa parent $P[u]$ wa node u n'imwe. Ahubwo, Madina akubwira M , umubare wa leaves, kandi ko index za leaves ari $0, 1, \dots, M - 1$. Task yawe ni ukugaragaza internal structure ya machine binyuze muburyo uyikoresha (operating).

Buri node ya machine ishobora gufata **ball** imwe gusa, kandi buri ball ifite **value** iri non-negative integer. Tuvuga ko node ifite value ya x iyo ifite ball ifite value ya x . Node iri **occupied** iyo ifite ball; bitabaye ibyo, iba iba iri **free**. Initially, buri node iba iri free.

Ushobora gukora ubwoko bubiri bwa operations:

- insert** (U, X) : Shyira value ya X kuri ball nshya hanyuma ugerageze kuyishyira kuri leaf ya U .
 - Iyo leaf U iri occupied, operation ireturninga `false` idashyizemo ball.
 - Iyo leaf U iri free, ball ishyirwa kuri node U . Hanyuma ball igenda yimukira kuri parent node mu gihe cyose iyo parent node ikiri free. Zino movements zihagarara iyo ball igeze kuri root cyangwa kuri node ifite parent iri occupied. Operation ireturninga `true`.
- collect()**: niba root iri free (bivuze ko machine iri empty), `collect` ireturninga array iri empty. Bitabaye ibyo, recursive procedure `traverse` isobanurwa hano hepfo iteranya values za balls zose ziri muri machine muri array S . Initially, array S iba iri empty ubundi `traverse(N - 1)` igahita iba called.

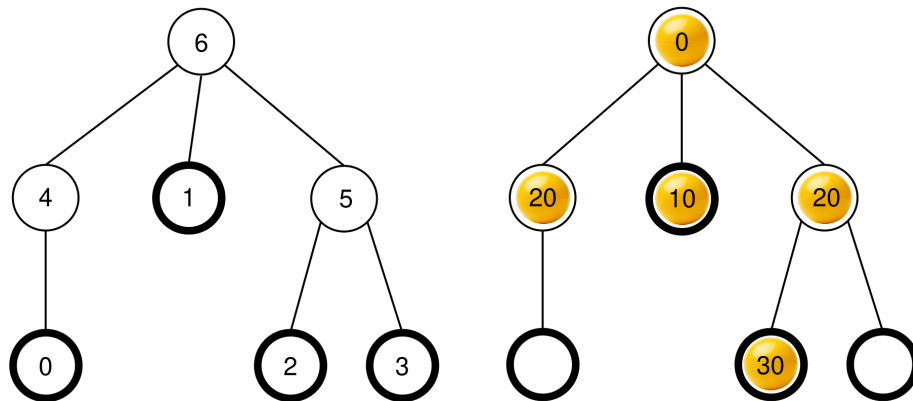
```

traverse(u):
    append the value of the ball at node u to the end of S
    let c[u] be the list of the occupied children of u
    sort c[u] in non-decreasing order according to the values
      of the balls at those nodes (if multiple nodes have the
      same value, they can appear in any order)
    for each v in c[u]:
        traverse(v)

```

Nyuma y'uko ino procedure irangiye, **balls zose zikurwa** muri machine, maze collect ika returninga S .

Urugero, fata machine ifite $N = 7$ nodes na parent array $P = [4, 6, 5, 5, 6, 6]$. Ifoto iri ibumoso igaragaza machine iri empty ifite node indices, mu gihe ifoto iri iburyo igaragaza configuration ishoboka ya balls nyuma ya sequence ya operations za `insert` (Ino sequence yerekanwa muri example section).



`collect()` niba called, root (node 6) iraba iri occupied, bityo S ibe initialized nka $S = []$ ubundi `traverse(6)` nayo ihite iba called.

traverse(6) : node 6 ifite value ya 0 ; kuyishyira kuri S bitanga $S = [0]$. Children za node 6 ni 4, 1, na 5, zose zikaba ziri occupied kandi values zazo ari 20, 10, na 20, uko bikurikirana. Gusortinga muri non-decreasing order bitanga $c[6] = [1, 5, 4]$ (menya ko $c[6] = [1, 4, 5]$ nabyo byaba bifite biri valid, kubera ko node 4 na 5 zifite value imwe). Hanyuma procedure icallinga `traverse` kuri buri node iri muri $c[6]$ muri ubwo buryo.

- **traverse(1)** : node 1 ifite value ya 10 ; kuyishyira kuri S bitanga $S = [0, 10]$. Node 1 nta children ifite, bityo ino call ihita irangira.
- **traverse(5)** : node 5 ifite value ya 20 ; kuyishyira kuri S bitanga $S = [0, 10, 20]$. Node 5 ifite one child irimo, node 2, bityo $c[5] = [2]$ ubundi procedure igacallinga `traverse(2)`.
 - **traverse(2)** : node 2 ifite value ya 30 ; kuyishyira kuri S bitanga $S = [0, 10, 20, 30]$. Node 2 nta children ifite, bityo ino call ihita irangira.

- Execution igaruka kuri `traverse(5)` yamaze guprocessinga children zose zo muri `c[5]`, bityo ino call ihita irarangira.
- **`traverse(4)`** : node 4 ifite value ya 20 ; kuyishyira kuri S bitanga $S = [0, 10, 20, 30, 20]$. Node 4 nta children ifite, bityo ino call ihita irangira.

Calls zose zarangiye kandi nta yandi ma node yo guprocessinga. Finally, buri ball yakuwe muri machine kandi `collect` yareturninze array $S = [0, 10, 20, 30, 20]$.

Task yawe ni ukumenya umubare wa amanode N no gutanga array of parents $R = [R[0], R[1], \dots, R[N-2]]$ yerekana uko machine iteye, ariko hakaba hari labeling iri distinct kuri nodes zitari leaf kandi zitari na root. Formally, array R utanga, ifatwa nkaho ari correct niba bishoboka guha label iri unique $L[u]$ aho $0 \leq L[u] < N$ kuri buri node u ya machine ku buryo:

- $L[u] = u$ kuri $0 \leq u < M$ na $u = N - 1$, kandi
- $L[P[u]] = R[L[u]]$ kuri $0 \leq u < N - 1$.

Ntabwo ari ngombwa ko $R[u] > u$. Igihe utanga solution yawe, **imashini igomba iri empty**.

Reka K ibe umubare wa operations za `collect` zakozwe, naho B ibe maximum value yanditswe kuri ball iyo ari yo yose muri calls zose za `insert`. Reka $C = K + B$. Hanyuma C **ntigomba kurenza 1000**. Amanota yawe muri subtasks zimwe na zimwe adendinga kuri value ya C .

Details Za Implementation

Ugomba ku implementinga procedure ikurikira.

```
std::vector<int> find_structure(int M)
```

- M : umubare wa leaf ziri muri machine.
- Procedure iba called rimwe kuri buri test case.
- Procedure igomba ku returninga array $R = [R[0], R[1], \dots, R[N-2]]$ ifite length $N - 1$, isobanura correct array ya parents.

Kugira ngo w'interactinge na machine, procedure yawe ishobora gucallinga izindi procedures ebyiri zikurikira.

Procedure ya mbere ni:

```
bool insert(int U, int X)
```

- U : index ya leaf yaho ball igomba gushyirwa. Condition $0 \leq U < M$ igomba kuba yujijwe.
- X : value ya ball. Condition $0 \leq X \leq 1000$ igomba kuzuzwa.
- Procedure ireturninga `true` iyo ball yashyizwemo, cyangwa `false` iyo leaf U yari iri occupied.

- Ino procedure ishobora kuba called inshuro zitarenze 500 000 .

Procedure ya kabiri ni:

```
std::vector<int> collect()
```

- Kusanya values za balls zose zinjijwemo, machine igasigara iri empty, hanyuma ikareturninga array S .

Niba igihe icyo ari cyo cyose muri execution yawe value ya C yarenze 1000 , solution yawe izabona verdict `Output isn't correct: Too many resources used'`.

Structure ya machine iba iri fixed mbere yuko `find_structure` iba called.

Constraints

- $2 \leq N \leq 1000$
- $1 \leq M \leq 200$
- $M < N$

Subtasks

Subtask	Score	Additional Constraints
1	5	The root has exactly M children.
2	10	$M \leq 3$
3	25	$N \leq 200, M \leq 45$
4	60	No additional constraints.

Muri subtasks 3 and 4, score yawe idependinga kuri value ya C mu buryo bukurikira.

Subtask 3 (25 points)

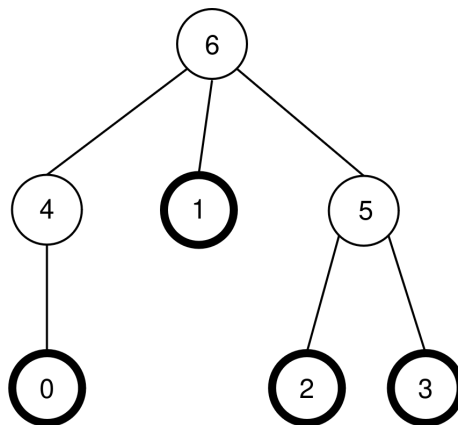
Limits	Score
$1000 < C$	0
$45 < C \leq 1000$	13
$C \leq 45$	25

Subtask 4 (60 points)

Limits	Score
$1000 < C$	0
$200 < C \leq 1000$	7
$71 < C \leq 200$	$47 - \frac{C}{5}$
$44 < C \leq 71$	$104 - C$
$C \leq 44$	60

Example

Considering a machine imwe nk'uko bigaragara muri description ya Task, bityo $N = 7$, $M = 4$, na $P = [4, 6, 5, 5, 6, 6]$. Imashini yongeye kugaragazwa ku ishusho ikurikira:



The grader makes the following call:

```
find_structure(4)
```

Uburyo bushobora gukora sequence ya calls:

1. **insert(0, 0)** : leaf 0 iri free, bityo ball ifite value 0 ishyirwa kuri leaf 0 . Node $P[0] = 4$ iri free, bityo ball yimukira kuri leaf 4 . Node $P[4] = 6$ iri free, bityo ball yimukira kuri leaf 6 . Kubera ko node 6 ari root, movements zirahagarara. Call ikareturninga `true` .
2. **insert(3, 20)** : leaf 3 iri free, bityo ball ifite value ya 20 ishyirwa kuri leaf 3 . Node $P[3] = 5$ iri free, bityo ball yimukira kuri node 5 . Kubera ko $P[5] = 6$ iri occupied, movements zirahagarara. Call ikareturninga `true` .
3. **insert(1, 10)** : leaf 1 iri free, bityo ball ifite value ya 10 ishyirwa kuri leaf 1 . Kubera ko $P[1] = 6$ iri occupied, ball iguma kuri node 1 . Call ikareturninga `true` .
4. **insert(2, 30)** : leaf 2 iri free, bityo ball ifite value ya 30 ishyirwa kuri leaf 2 . Kubera ko $P[2] = 5$ iri occupied, ball iguma kuri node 2 . Call ikareturninga `true` .

5. **insert**(1, 25) : leaf 1 iri occupied, bityo ball ntishyirwa kuri leaf 1 . Call ireturninga `false` .
6. **insert**(0, 20) : leaf 0 iri free, bityo ball ifite value ya 20 ishyirwa kuri leaf 0 . Node $P[0] = 4$ iri free, bityo ball yimukira kuri node 4 . Kubera ko $P[4] = 6$ iri occupied, movements zirahagarara. Call ikareturninga `true` .

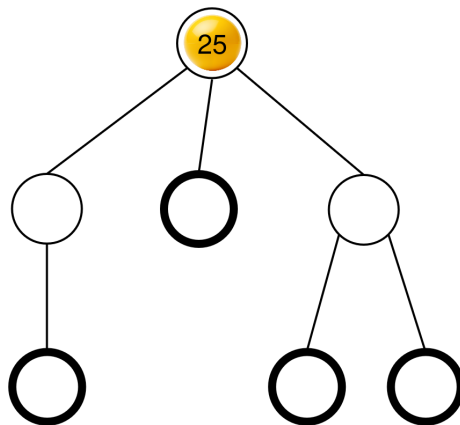
Configuration yanyuma ya balls yari yerekanywe kari muri task description.

Hanyuma, procedure igacallinga:

```
collect()
```

Execution y'ino call nayo yari yasobanuwe muri task description, bityo ino call izareturninga $S = [0, 10, 20, 30, 20]$. Nyuma, machine yongere kuba empty.

Hanyuma, procedure igacallinga **insert**(2, 25) . Leaf 2 iri free, bityo ball ifite value ya 25 ishyirwa kuri leaf 0 . Ino ball ihita yimukira kuri node 5 , hanyuma ikajya kuri node 6 , aho ihagarara. Call ikareturninga `true` . Configuration ya balls ivuyemo irerekanywa mugishushanyo gikurikira:



Hanyuma, procedure igacallinga **collect**() ikareturninga $S = [25]$ hanyuma machine igahita iba empty.

Hari correct parent arrays ebyiri `find_structure` ishobora ku returning: $R = P = [4, 6, 5, 5, 6, 6]$ (ijyanye na labels $L = [0, 1, 2, 3, 4, 5, 6]$), na $R = [5, 6, 4, 4, 6, 6]$ (ijyanye na labels $L = [0, 1, 2, 3, 5, 4, 6]$). Muri uru rugero, $K = 2$ na $B = 30$, bityo $C = 32$.

Sample Grader

Input format:

```
N M
P[0] P[1] P[2] ... P[N-2]
```

Output format:

```
K B C
Q
R[0] R[1] R[2] ... R[Q-1]
```

Aha, Q ni length ya array R yabaye returned na `find_structure`.